

Mohamed Afrath Segu Mohamed

Full-Stack Software Engineer · C#/.NET Core, REST APIs & Auth · Angular/TypeScript · SQL Server
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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 3+ years of experience in developing and maintaining production applications across backend, database and frontend layers. Proficient in C#, ASP .NET Core, SQL Server and Angular. Experienced in designing and securing multi tenant APIs using JWT and role based access control, and optimizing SQL Server queries to reduce reporting time by up to 60%. Hands on in complete development life cycle from backend services and data layers to reactive Angular interfaces. Holds an MSc in Computing (Artificial Intelligence) from Dublin City University, with practical projects on LLM, multi agent systems, semantic image retrieval and automated test scenario generation.

TECHNICAL SKILLS

Backend & APIs: C#, .NET Core, .NET Framework, ASP.NET Core, REST API design, Microservices, Entity Framework Core, LINQ
Databases: MS SQL Server, Query Optimisation, Stored Procedures, Performance Tuning
Frontend: Angular, TypeScript, JavaScript, RxJS, React, HTML5, CSS
Auth & Security: JWT, RBAC, Claims-Based Identity, API authorisation
AI / LLM Tooling: LLMs, Prompt Engineering, NLP, Hugging Face Inference API, Transformer models (T5, Qwen, Nemotron), BLIP, BM25/VSM information retrieval, Flask, Python
DevOps & Cloud: Git, Azure DevOps Pipelines, GitHub Actions, Docker, Kubernetes, Azure (Service Bus, Blob/Queue Storage)
Agile: Scrum (stand-ups, retrospectives), Azure DevOps Boards, distributed/cross-team collaboration

PROFESSIONAL EXPERIENCE

Software Developer <i>TVS Next · Chennai, India</i>	Jun 2021 – Aug 2024
- Delivered full-stack features across two production systems: a banking modernisation platform for a US client serving 5000+ users and an internal Employee Management System supporting 500+ employees. - Developed and maintained 10+ RESTful APIs within a microservices architecture, applying SOLID and clean architecture principles, and secured them with JWT, RBAC, and claims-based identity for a multi-tenant financial environment. - Optimised MS SQL Server through stored procedure improvement, schema optimisation and query tuning, cutting average query execution time 40–60% on reporting operations. - Built reusable Angular component libraries, reactive forms, and data tables using RxJS, and held ~60–70% test coverage with Karma/Jasmine and xUnit/Moq under TDD across a 10+ developer team. - Implemented Azure Service Bus and Blob/Queue Storage for event-driven workflows; containerised services with Docker and built Azure DevOps CI/CD pipelines, cutting environment setup from 30–60 min to under 10. - Collaborated directly with stakeholders in an Agile/Scrum process to turn business requirements into technical solutions; mentored teammates via code review and received three performance awards (Spotlight of Excellence 21-22 & 23-24, Super Squad Award 23-24).	

PROJECTS

Automated Gherkin Test Scenario Generation Using Transformer Models <i>MSc Research — DCU, 2025</i>
Developed an NLP system using T5, Qwen, and Nemotron to auto-generate Gherkin test scenarios from user stories via NLP and structured prompt engineering; reached a BERTScore F1 of 0.834, validated with BLEU, ROUGE-L, LLM-as-a-Judge and human review.
Clash of Minds AI (Multi-Agent AI Debate & LLM Judge) <i>Personal Project - 2025</i>
Developed a multi-agent system where two LLMs debate a user-chosen topic and a third model judges the arguments and picks a winner, using JavaScript, Hugging Face Inference API, LLMs, Prompt Engineering.
AI Vista (AI-Powered Semantic Image Retrieval System) <i>Personal Project - 2025</i>
Developed a semantic image search engine combining BLIP-generated captions with classic IR like VSM, BM25, and unigram language model ranking, using Python, Flask, BLIP, BM25, VSM, Scrapy.

EDUCATION

MSc in Computing (Artificial Intelligence) — <i>Dublin City University, Ireland</i>	2024 – 2025
B.E. in Computer Science and Engineering — <i>Velammal Engineering College, India</i>	2017 – 2021

CERTIFICATIONS

Azure Developer Associate (Microsoft, 2023) · AZ-900 Azure Fundamentals (Udemy, 2023) · MS SQL Server Masterclass (Udemy, 2023) · Angular & ASP.NET Core REST API (Udemy, 2022) · Angular & C# Basics (HackerRank, 2021)